

The project was developed using a set of powerful Python-based tools and libraries for data processing, machine learning, visualization, and deployment. These technologies provide the foundation for efficient model development, analysis, and application deployment.

****Anaconda Navigator:**** A desktop graphical user interface used to launch applications and manage Python packages and environments for data science.

<<https://www.anaconda.com/download>>

****PyCharm:**** An Integrated Development Environment (IDE) specifically designed for Python programming, offering code analysis and debugging tools.

<<https://www.jetbrains.com/pycharm/>>

****NumPy:**** Core library for numerical computing in Python, providing support for large, multi-dimensional arrays and mathematical functions. <<https://numpy.org/doc/stable/>>

****Pandas:**** Data manipulation and analysis library offering data structures and operations for manipulating numerical tables and time series.

<<https://pandas.pydata.org/docs/>>

****Scikit-learn:**** Machine learning library featuring classification, regression, and clustering algorithms like support vector machines and random forests. <<https://scikit-learn.org/stable/>>

****Matplotlib:**** Comprehensive library for creating static, animated, and interactive visualizations in Python. <<https://matplotlib.org/stable/>>

****Seaborn:**** Data visualization library based on matplotlib that provides a high-level interface for drawing attractive statistical graphics.

<<https://seaborn.pydata.org/>>

****Flask:**** A lightweight WSGI web application framework used to build web APIs and deploy machine learning models. <<https://flask.palletsprojects.com/>>